Write Modules and Save Lives

Part—1

OpenMRS, a widely used Electronic Medical Record system, is a community-developed, open source and enterprise grade platform. Its core is very thin, and almost every functionality is provided through pluggable modules. It is robust enough for a nationwide healthcare system and nimble enough for a field-based clinic. This article series will cover OpenMRS module development; it is aimed at J2EE developers and those who want to contribute to OpenMRS.

There are a lot of OpenMRS modules and the list is constantly increasing. A detailed list is available at *https://modules.openmrs.org*. OpenMRS offers the functionality to directly install or update modules through a Web interface—on the fly, without restarting Tomcat.

This article will cover the basics of the OpenMRS architecture and how to install, update, stop and start a module in OpenMRS. The OpenMRS is built on the Spring and Hibernate stack and uses MySQL as the database.

Basic architecture

The components are:

- Spring: An application development framework for developing MVC architecture applications
- POJO: Plain Old Java Object
- Hibernate: The Java persistence framework project, which performs powerful object-relational mapping, and queries databases using HQL and SQL
- DAO: Direct Access Object
- Liquibase: Database-independent library for tracking, managing and applying database changes

An interesting fact about OpenMRS modules is that they are like mini subsystems that can not only communicate with OpenMRS core services, but can also have their own services and schema that can be used by other modules. This makes modules flexible enough to be independent of the system, and robust enough to be dependent on each other, if required.

Module installation

> **Note:** Installing modules through the Web interface is disabled by default (for security purposes). To enable it, set the *'module.allow_web_admin'* runtime property to *true* and restart the application.

1. Download the module to the local filesystem.